HACETTEPE UNIVERSITY
DEPARTMENT OF COMPUTER ENGINEERING
CMP422 Computer Networks
Homework 2

Subject: Socket Programming
Programming Language: Python
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Deadline: 27.10.2026 – 23:59

Introduction / Aim

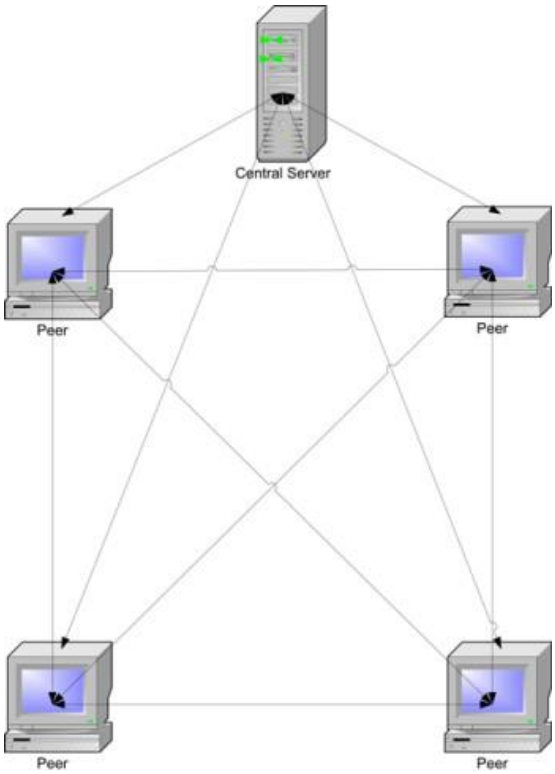
The aim of this experiment is to introduce you the basic concepts of socket programming on python. You are expected to build a peer-to-peer messaging and file exchange software on TCP/IP protocol stack by using network programming tools of UNIX and python programming language. Within this context you will explore how to use sockets and threads in python. The software that you are going to develop should include two components: a server and a client.

Experiment

You will develop a text messaging and file sending application. The system is composed of a central server and clients (peers). The server will store information about connected peers and keep a list of online peers (userlist). The userlist contains user names, IP addresses, and port numbers of online peers. The server sends the userlist to all online clients in order to allow them to exchange text messages and files among themselves.

When a peer connects to the server, it sends its username, IP address, and port number (port number will be randomly selected by client from available port numbers) to server and then gets the userlist. After getting the userlist from the server, the peer can send text messages or files to other online peers. Message and file sending operations will be carried on “peer-to-peer” basics. There will be no communication through server when exchanging messages and files. User list will be sent all online users when a new user connects.

While designing your project, be aware that the IP addresses and port numbers of peers may change in every sign-in operation. In other words, peers may connect to the server with different IP addresses and port numbers. When a peer connects to the server, an updated userlist will be sent to all online peers. You should use threads to cope with multiple connections. A sample network schema of the system is shown below:



Usage

Server: The server can run on any port number available. The server port number will be provided from the command line as the first parameter. The clients will also use this port number when connecting to the server. The server logs all of its operations. Log formats are very strict as they will constitute an important part of the evaluation process. Sample log files for the server is as follows:

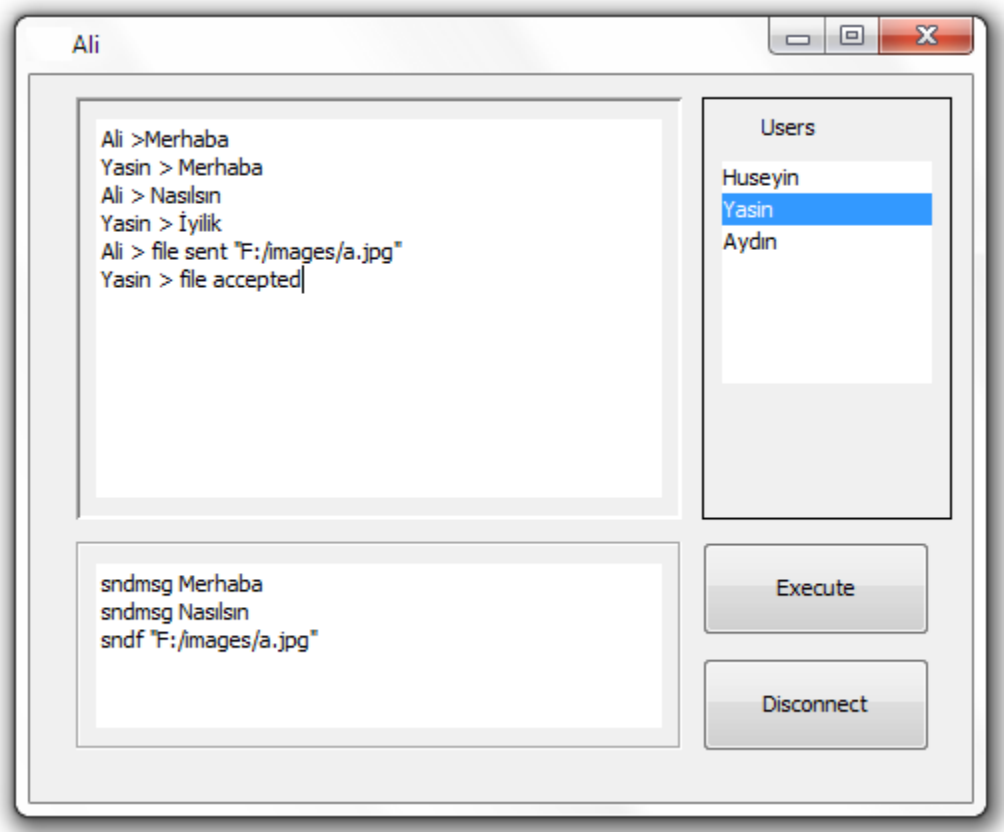
Server log

193.146.234.78 1020 connected
193.146.236.88 1021 connected
193.146.236.88 1021 disconnected

Client: The client software will provide a GUI to exchange files and messages among clients. You can use python components to implement GUI of the client program. The client application will take IP address and port number of the server and username from the command line. It is guaranteed that a unique username will be provided for each client while testing. The command format for the client program is as follows:

python client <server-ip-address> <server-port-number> <user-name>

The GUI will contain a list to show users that are currently online, a text area to use as command window and a text area to show incoming messages from selected user. The user will enter commands in the command textbox of the GUI to send messages or files. Results of these operations will be printed in the output area. User will select a user from user list that lies at the right side of the window. After selecting the remote user, you can send messages or files. Your outgoing and incoming messages will be printed in the output text. A sample GUI is shown below:



Commands will be executed after the user presses execute button. User can execute multiple commands at once.

The client program logs all of its operations. A sample log file for the client is given below:

Client log

Message to Ali@193.146.234.78:2020 “Merhaba”
a.jpg to Aydın@193.146.234.88:1021
Message from Huseyin@193.146.234.98:3013 “Nasılsın”
b.jpg from Yasin@193.146.234.28:1234

Command format for clients

A user of client program can send messages to other users by using sndmsg command. All incoming messages will be shown in the output area. The format of commands is as follows:

sndmsg <message content>

example:
sndmsg Merhaba nasilsin

sndf <full-path-of-the-file>

example:
sndf "F:/resimler/a.jpg"

All incoming files will be stored in a download directory, which is in the current working directory of the client program. When a file is sent to another client, a message box will be displayed at the destination client's GUI to confirm operation. When the user confirms file sending (upload) operation, file transfer will be started. A user can continue to send messages while a file download/upload is continuing.

Last Remarks:

- Regardless of the length, use UNDERSTANDABLE names to your variables, classes and functions.
- Write READABLE SOURCE CODE block
- You will use online submission system to submit your experiments.
https://submit.cs.hacettepe.edu.tr
- You can ask your questions through course's Piazza Group and you are supposed to be aware of everything discussed in the newsgroup.